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Effects of partial neuromuscular blockade on lateral spread response monitoring during microvascular decompression surgery ☆

Yang Hoon Chung [a,1], Won Ho Kim [b,1], Ik Soo Chung [a], Kwan Park [c], Seong Hyuk Lim [d], Dae Won Seo [d], Jeong Jin Lee [a,*], Song-I Yang [a]

[a] Department of Anaesthesiology and Pain Medicine, Samsung Medical Centre, Sungkyunkwan University School of Medicine, Seoul, Republic of Korea
[b] Department of Anaesthesiology and Pain Medicine, Samsung Changwon Hospital, Sungkyunkwan University School of Medicine, Changwon, Republic of Korea
[c] Department of Neurosurgery, Samsung Medical Centre, Sungkyunkwan University School of Medicine, Seoul, Republic of Korea
[d] Department of Neurology, Samsung Medical Centre, Sungkyunkwan University School of Medicine, Seoul, Republic of Korea

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HIGHLIGHTS

- The efficacy of partial neuromuscular blockade (NMB) for intraoperative monitoring of lateral spread responses (LSRs) has not been established yet.
- The successful intraoperative LSR monitoring can be achieved under partial NMB with train-of-four and T1 amplitude monitoring.
- The spontaneous free-run facial EMG occurred less frequently in patients with partial NMB.

ABSTRACT

Objective: We evaluated the effect of partial neuromuscular blockade (NMB) and no NMB on successful intraoperative monitoring of the lateral spread response (LSR) during microvascular decompression (MVD) surgery.
Methods: Patients were randomly allocated into one of three groups: the TOF group, the NMB was targeted to maintain two counts of train-of-four (TOF); the T1 group, maintain the T1/Tc (T1: amplitude of first twitch, Tc: amplitude of baseline twitch) ratio at 50%; and the N group, no relaxants after tracheal intubation. Successful LSR monitoring was defined as effective baseline establishment and maintenance of the LSR until dural opening.
Results: The success rate of LSR monitoring was significantly lower in the TOF group. But, there was no significant difference between T1 and N. The detection rate of spontaneous free-run electromyography (EMG) activity was significantly higher in the N group compared with the TOF and T1 groups.
Conclusions: Partial NMB with a target of T1/Tc ratio at 50% allows good recording of LSR with same outcome as surgery without NMB, and reduced spontaneous EMG activity.
Significance: We suggested the availability of partial NMB for intraoperative LSR monitoring.
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1. Introduction

Hyperactive dysfunction of facial nerves can cause a hemifacial spasm (HFS), which is characterized by chronic and involuntary contraction of the ipsilateral portion of the face (Miller and Miller, 2012; Sindou, 2005; Yamashita et al., 2005). Spontaneous recovery from an HFS is very rare; moreover, HFSs can lead to serious aesthetic and functional disabilities (Neves et al., 2009).

Primary HFSs are generally related to vascular compression of the facial nerve at the root exit zone (Miller and Miller, 2012; Sindou, 2005). Several treatment options for primary HFSs have

☆ This study was conducted in Samsung Medical Centre, Sungkyunkwan University School of Medicine, Seoul, Republic of Korea.

* Corresponding author at: Department of Anaesthesiology and Pain Medicine, Samsung Medical Centre, Sungkyunkwan University School of Medicine, 50, Irwon-Dong, Gangnam-Gu, Seoul 135-710, Republic of Korea. Tel.: +82 2 3410 0784; fax: +82 2 3410 6626.
E-mail address: jjeong.lee@samsung.com (J.J. Lee).

[1] Y.H. Chung and W.H. Kim contributed equally to this article.

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been proposed. Oral medications, such as anticonvulsants, are sometimes prescribed for mild HFSs. However, the use of these medications is often hindered by their poor effectiveness and possible side effects (Miller and Miller, 2012). Another option is to administer injections of botulinum toxin, which achieves temporary treatment; however, this effect generally persists for only 3–4 months (Costa et al., 2005; Miller and Miller, 2012; Wang et al., 2013). The only known curative treatment to date is microvascular decompression (MVD) surgery (Miller and Miller, 2012; Sindou, 2005).

Hyperexcitable facial nerves can cause an abnormal muscle response known as the lateral spread response (LSR) in patients with a primary HFS (Fernandez-Conejero et al., 2012; Sindou, 2005; Yamashita et al., 2005). The LSR can be detected by electromyography (EMG) monitoring of the facial muscles upon stimulation of another branch of facial nerves (Fernandez-Conejero et al., 2012; Moller and Jannetta, 1985b; Sindou, 2005). Møller and Jannetta reported that the LSR disappears immediately upon removal of the offending vessels and if it remains at the end of operation, the spasms are more likely to persist after surgical procedure (Moller and Jannetta, 1985a,b, 1987). Accordingly, intraoperative LSR monitoring is considered to be an effective diagnostic tool for complete decompression and also to be an important prognostic factor for MVD surgery (Kong et al., 2007; Sekula et al., 2009; Yamashita et al., 2005).

Anesthetic drugs, especially neuromuscular blocking agents, can significantly influence the success of EMG monitoring (Kizilay et al., 2003; Sloan, 2013). Previous studies have recommended the avoidance of neuromuscular blockade (NMB) during intraoperative electrophysiologic monitoring (IOM) (Kim et al., 2013; Kothbauer, 2007). However, some neurosurgeons feel uncomfortable about avoiding NMB, considering the risks of patient movement during surgery and/or patient movement by transcranial stimulation. In some cases, large doses of anesthetics are needed for complete immobility; such doses can cause hemodynamic instability in some patients (Cai et al., 2009; Kizilay et al., 2003). Several pieces of evidence support the idea that successful IOM can be conducted under partial NMB and patient immobility (Kizilay et al., 2003; Lennon et al., 1992; Sloan, 2013). For example, previous studies have reported successful EMG monitoring of facial nerves in otologic surgeries (Cai et al., 2009; Ho et al., 1989; Kizilay et al., 2003; Lennon et al., 1992; Sloan, 2013); additional studies have found that facial muscles have a lower sensitivity to NMB compared with other muscle types (Abdulatif and el-Sanabary, 1997; Rimaniol et al., 1996). Furthermore, spontaneous tonic contraction of facial muscles in patients with a high tonic grade (Lee et al., 2012) can prevent LSR monitoring; thus, partial NMB could be beneficial to these patients. Therefore, LSR monitoring during MVD surgery may be possible with partial NMB, and LSR monitoring under partial NMB may be preferable to LSR monitoring in some patients. However, no clinical studies have yet compared the success of LSR monitoring under partial NMB with that under NMB, or compared the effectiveness of different degrees of partial NMB for LSR monitoring. Therefore, the current study was designed to compare two different degrees of partial NMB, in addition to the absence of NMB, on the success of LSR monitoring in MVD surgery.

2. Methods

This prospective, randomized, controlled study was approved by the Hospital Institutional Review Board. This study is registered at the www.clinicaltrials.gov website, under protocol number NCT01598961, May 2012.

One hundred fifty patients who were diagnosed with an HFS and who underwent MVD surgery with intraoperative LSR monitoring between May 2012 and February 2013 were enrolled in this study (Fig. 1). Patient age at the times of surgery ranged from 20 to 70 years. Patients with renal diseases, neuromuscular diseases, and American Society of Anesthesiologists (ASA) physical status class III or more were excluded from this study.

After obtaining written informed consent from each patient, patients were randomly allocated into one of three groups, in a 1:1:1 allocation ratio. Randomization was conducted using random numbers generated by internet based computer program (www.randomizer.org). It created 1 set of 150 non-unique numbers ranging from 1 to 3 (matched up 1 with TOF group, 2 with T1 group, and 3 with N group, respectively). All attending neurophysiologists and surgeons were blinded to the study group assignments of the patients. All surgical procedures were performed by the same neurosurgeon (K. Park).

2.1. Anesthesia and research protocol

General anesthesia was induced and maintained with propofol and remifentanil. Using supramaximal stimuli, the baseline T1 amplitude (Tc) was established using a neuromuscular transmission module (M-NMT module, Datex-Ohmeda Inc., Helsinki, Finland) (Dahaba et al., 2002). This module measures the response of the adductor pollicis brevis muscle, using train-of-four (TOF) stimulation of the ulnar nerve, and visually displays the associated TOF count, ratio, and T1/Tc value (Fig. 2). Rocuronium (0.6 mg/kg) was infused to facilitate tracheal intubation. Propofol and remifentanil were continuously infused through a target controlled infusion (TCI) pump (Orchestra™, Fresenius Vial, France). Propofol (3–5.5 μg/ml) was adjusted to maintain a BIS level of 40–60. Remifentanil (1–4 ng/ml) was titrated to control patient arterial blood pressure within a 20% range of its preoperative value.

Subjects were randomly allocated to one of three groups. In the TOF group, NMB was targeted to maintain two counts of TOF during the LSR monitoring. In the T1 group, NMB was targeted to maintain a T1/Tc ratio of 50% during the LSR monitoring (Fig. 2). In the N group, no additional NMB drugs were used after tracheal intubation. In the TOF and T1 groups, an intravenous vecuronium continuous infusion (0.5 μg/kg/min) was initiated when the appropriate NMB target was reached after tracheal intubation. During surgery, the TOF response of the adductor pollicis brevis muscle was monitored every 5 min with the NMT module; the vecuronium dose was also adjusted every 5 min to maintain the appropriate NMB target.

The primary endpoint of this study was the incidence of successful LSR monitoring. We assessed the success of LSR detection at six time points: (1) before skin incision (baseline), (2) before dural opening, (3) after dural opening, (4) before decompression, (5) after decompression, and (6) after dural closure. In some patients, the LSR would disappear with dural opening or after drainage of the cerebrospinal fluid (CSF) before decompression (Kim et al., 2010). Accordingly, the success of LSR monitoring was defined as the successful detection and establishment of the baseline LSR, and the maintenance of this LSR before dural opening or decompression. Failure to detect the LSR before dural opening (time points 1 and 2) was regarded as a monitoring failure. Reduction of the EMG amplitude by greater than 50% of the baseline as reported by the neurophysiologist, without a surgical reason, was also considered to be a monitoring failure.

The secondary outcome variables of this study included the incidence of spontaneous patient movements during EMG monitoring, the incidence of spontaneous free-run EMG activity, the mean doses of propofol–remifentanil, and the mean BIS level. The doses of propofol–remifentanil and BIS levels were recorded

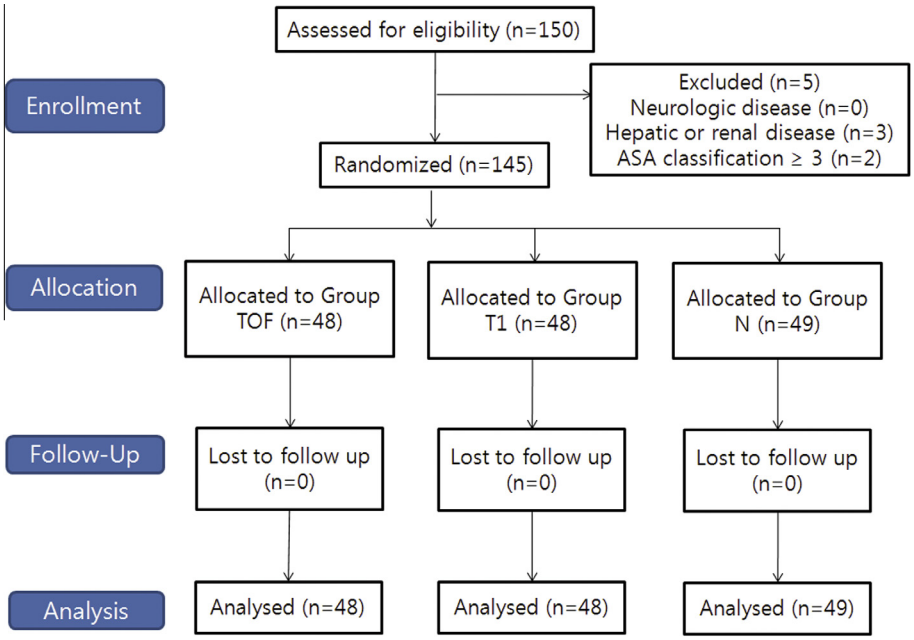

Fig. 1. Flow diagram of the study procedure, according to the CONSORT 2010 guidelines.

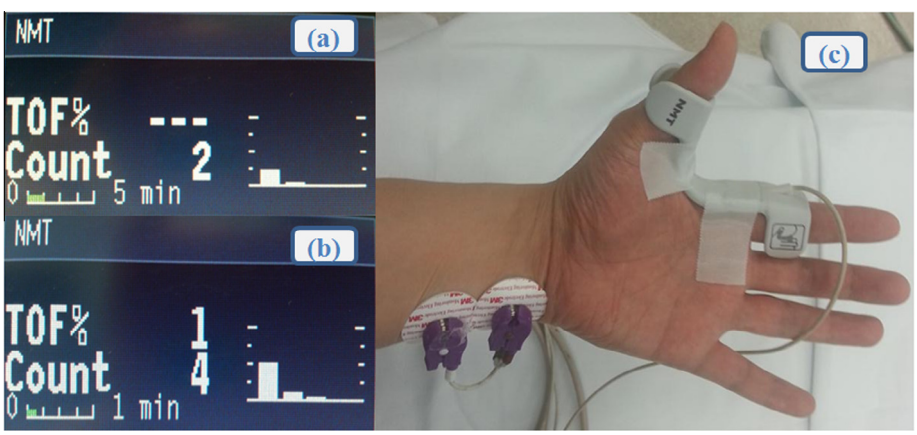

Fig. 2. Images of neuromuscular blockade (NMB) monitoring using NMT module. The target of two counts of train-of-four (TOF) in TOF group (a), T1/Tc ratio of 50% in T1 group (b), and set up image (c).

every 15 min during LSR monitoring. All mean values reported in this study were calculated by averaging all recorded variables.

Patients were routinely followed up at one week, three months, and one year after discharge. We considered absent of spasm as complete relief, >50% decrease in the initial spasm as partial relief, and <50% or persisted spasm as no relief. This clinical outcome was compared among groups at three follow up.

2.2. Recording EMG

Stimulating needle electrodes were inserted intradermally over the zygomatic and buccal branches of the facial nerve. EMG signals were recorded with subdermal needle electrodes from the frontalis, orbicularis oculi, orbicularis oris, and mentalis muscles (Fig. 3) using a Protektor IOM device (XLTEK, Oakville, Canada). Single 0.2-ms pulse waves with intensities ranging from 1 to 30 mA were used for stimulation. We were able to record the LSR of the frontalis and orbicularis oculi muscles by stimulating the buccal branch, and were also able to record the LSR of the orbicularis oris and mentalis muscles by stimulating the zygomatic branch (Fig. 4). A neurophysiologist monitored the free-run EMG signals throughout the duration of the main surgery (Minahan and Mandir, 2011).

2.3. Statistical analysis

The sample size of this study was determined based on our pilot study. The success rate of LSR monitoring, under the target TOF count of two, was approximately 50% in our pilot study. We hypothesized that the no-NMB group (the N group) and the partial NMB groups (the TOF and T1 groups) would exhibit more than 30% differences in their success rates. Power analysis revealed that 47 patients would be required in each group ($\alpha = 0.05$, $\beta = 0.20$). Thus, we enrolled 50 patients in each group, in order to allow for the possibility of dropouts.

Statistical analyses were conducted using SAS, version 9.3 (SAS Institute, Cary, NC, USA). For continuous variables, one way ANOVA was used when the variables were normally distributed; otherwise, the Kruskal–Wallis test was used. Tukey's test was used for

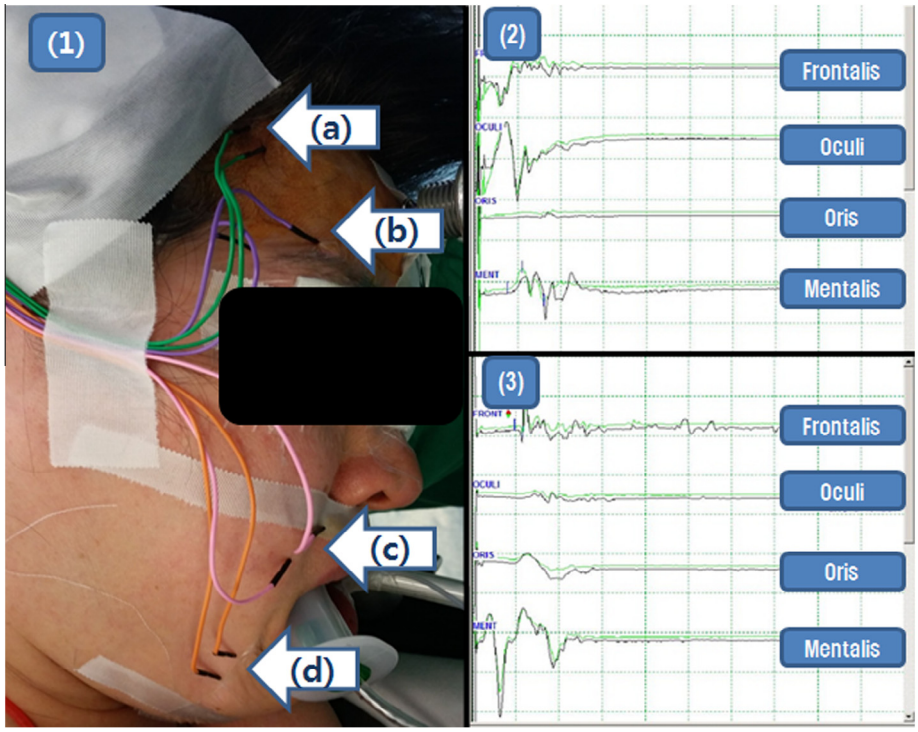

Fig. 3. Set up images for lateral spread response (1). Stimulating needles are inserted in temporalis (a), orbicularis oculi (b), orbicularis oris (c), and mentalis (d) muscles. Baseline electromyogram upon stimulation of zygomatic branch (2), and buccal branch (3).

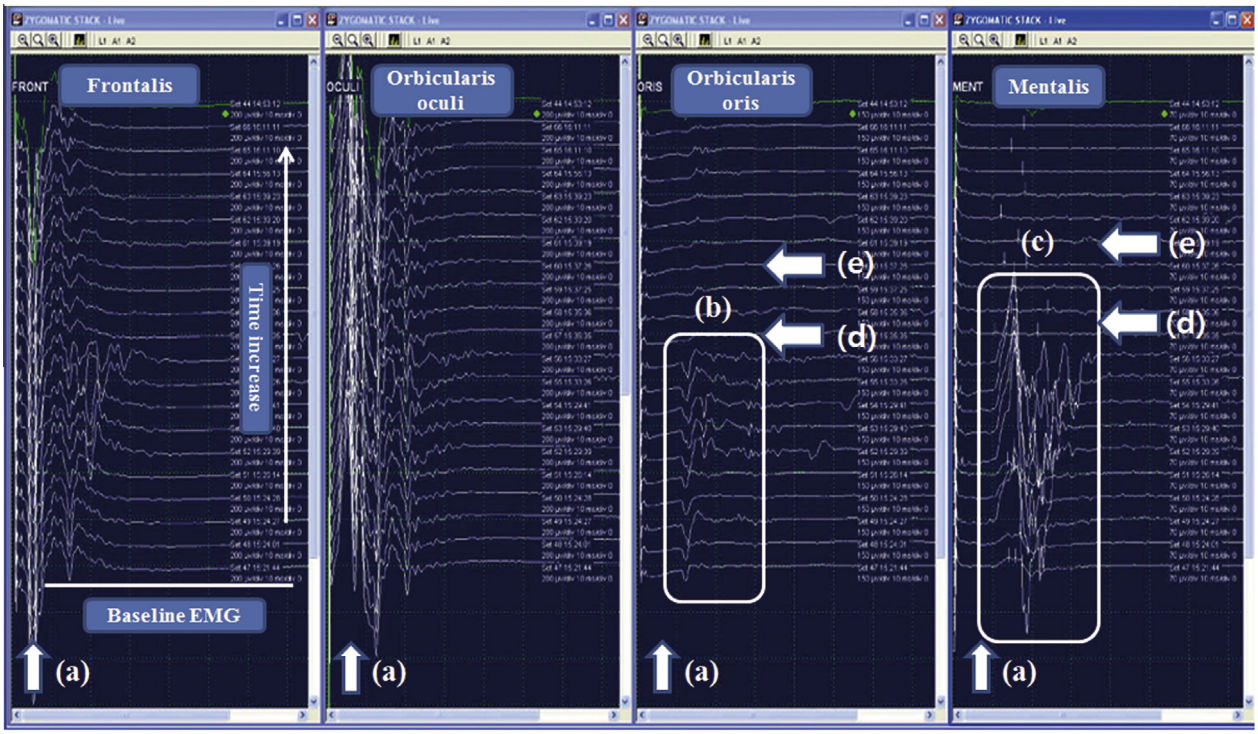

Fig. 4. Stack wave image of an intraoperative facial electromyogram (EMG) upon stimulation of the zygomatic branch. Electrical stimulation of the zygomatic branch (a) produced an abnormal muscle response, called the lateral spread response (LSR), in the orbicularis oris (b) and mentalis (c) muscles. This abnormal response disappeared (e) after decompression (d).

post hoc analysis of continuous variables. For categorical variables, either the chi-square test or Fisher's exact test was used. The Bonferroni method was used for multiple testing of categorical variables. *P*-values less than 0.05 were considered significant.

3. Results

One hundred fifty patients were enrolled and considered for eligibility. Five patients were removed from the study upon appli-

cation of our exclusion criteria (Fig. 1). Patient demographic and preoperative data are shown in Table 1; no significant differences were observed between any of the three groups. The rates of previous botulinum toxin use were not different among any of the groups (20.83%, 25%, and 34.69% in the TOF, T1, N groups, respectively, $P = 0.286$).

Patient intraoperative parameters are shown in Table 2. The mean BIS levels, mean potassium levels, and mean surgery times did not differ between any of the three groups (Table 2). Similarly, the mean propofol TCI doses and mean BIS levels did not differ between any of the groups. However, the mean remifentanil TCI dose was significantly higher in the N group compared with the other two groups (1.61, 1.70, and 2.1 ng/ml in the TOF, T1, and N groups, respectively, $P < 0.001$). The success rate of LSR monitoring was significantly lower in the TOF group than in the T1 and N groups (66.67%, 91.67%, and 95.92% in the TOF, T1, and N groups, respectively, $P < 0.001$). The detection rate of spontaneous free-run EMG activity was significantly higher in the N group than in the TOF and T1 groups (6.25%, 6.25%, and 28.57% in the TOF, T1, and N groups, respectively, $P = 0.001$). Maintenance of LSR at the end of surgery was observed in 5 patients ($n = 1$, 1, and 3 in the TOF, T1, and N groups, respectively, $P = 0.871$).

For four patients in the N group, the neurophysiologist requested a higher level of NMB due to interference in the LSR monitoring because of spontaneous free-run EMG activity; thus, for these patients we initiated a continuous infusion of vecuronium and aimed for an initial T1/Tc ratio of 50% in the NMB.

Spontaneous patient movement during surgery occurred in six patients (coughing, 3 cases; slight movements of the lower extremities, 3 cases). The incidences of patient spontaneous movement were not significantly different between any of the three groups ($n = 2$, 1, and 3 in the TOF, T1, and N groups, respectively, $P = 0.871$) (Table 2). All movements occurred during a position change of the patient, or during a surgical incision. No incidences of movement during microscopic surgery were noted.

There was no significant difference in clinical outcomes between the three groups (Table 3). At 1 year follow up, the rate of complete relief among those in whom LSR disappeared was 26/34 (76.47%), 27/37 (72.97%), 28/35 (80%) in the TOF, T1, and N groups, respectively ($P = 0.782$, the patients who were lost to follow up excluded). There were no significant nerve injuries or adverse events in all groups.

4. Discussion

We compared the success rates of LSR monitoring with three different levels of NMB, in addition to the absence of NMB. Sloan et al. suggested that intraoperative EMG monitoring is possible using partial NMB targeted to a TOF count of two, or a T1/Tc ratio of 10–20% (Sloan, 2013; Sloan and Heyer, 2002); however, our pilot study revealed that the success rate of LSR monitoring was quite low (approximately 50%) at these targets. Although it is generally accepted that NMB is not required during motor-evoked potential monitoring, this assumption is not yet supported by clinical trial-based evidence showing that the absence of NMB is better for LSR monitoring, compared with partial NMB.

We referred to the literature to determine the optimal target of partial NMB. As mentioned above, either a TOF count of two or a T1/Tc ratio of 10–20% can be used for intraoperative EMG monitoring (Sloan, 2013; Sloan and Heyer, 2002). However, some clinicians

Table 1
Comparison of patient preoperative data between the three groups.

Variable	TOF group ($N = 48$)	T1 group ($N = 48$)	N group ($N = 49$)	*P*-value
Sex (M/F)	16/32	16/32	12/37	0.549
Age (year)	52.1 (10.29)	50.44 (11.78)	55.91 (18.65)	0.330
Weight (kg)	61.49 (11.10)	64.48 (11.71)	60.94 (9.34)	0.276
Height (cm)	161.24 (8.57)	160.75 (7.97)	159.59 (7.28)	0.598
BMI	23.55 (3.03)	24.91 (3.98)	23.91 (3.11)	0.131
Previous anticonvulsant use (*n*, %)	2 (4.17%)	3 (6.25%)	2 (4.08%)	0.898
Preoperative botulinum toxin use (*n*, %)	10 (20.83%)	12 (25%)	17 (34.69%)	0.286
Preoperative tonic grade (I/II/III/IV)	5/20/20/3	2/18/19/9	3/23/18/5	0.534

*Values are presented as mean (SD).
†BMI: body mass index.

Table 2
Comparison of intraoperative parameters.

Variable	TOF group ($N = 48$)	T1 group ($N = 48$)	N group ($N = 49$)	*P*-value
Mean propofol dose (TCI dose, μg/ml)	3.7 (0.4)	3.7 (0.4)	3.9 (0.5)	0.066
Mean remifentanil dose (TCI dose, ng/ml)	1.61 (0.61)[N]	1.70 (0.52)[N]	2.1 (0.70)[TOF,T1]	<0.001*
Mean BIS	38.01 (5.7)	37.22 (2.67)	36.31 (4.19)	0.409
Mean potassium level (mmol/L)	3.52 (0.26)	3.44 (0.17)	3.52 (0.22)	0.653
Surgery time (min)	114 (17)	118 (17)	110 (13)	0.087
Mean vecuronium dose (μg/kg/min)	0.62 (0.15)	0.44 (0.13)	–	<0.001*
Incidence of spontaneous movement (*n*, %)	2 (4.17%)	1 (2.08%)	3 (6.12%)	0.871
Success rate of LSR monitoring (*n*, %)	32 (66.67%) [T1,N]	44 (91.67%) [TOF]	47 (95.92%)[TOF]	<0.001*
Detection of spontaneous free-run EMG activity (*n*, %)	3 (6.25%)[N]	3 (6.25%)[N]	14 (28.57%)[TOF,T1]	0.001*
Maintenance of LSR at the end of surgery (n,%)	1 (2.08%)	1 (2.08%)	3 (6.12%)	0.871

†TCI: target controlled infusion, LSR: lateral spread response, EMG: electromyography.
‡The mean vecuronium infusion dose is the average of all continuous infusion doses every 15 min during the maintenance of partial NMB.
[TOF] Significantly different from the TOF group.
[T1] Significantly different from the T1 group.
[N] Significantly different from the N group.
* Values are presented as mean (SD), *n* (%).

Table 3
Comparisons of clinical outcomes at 1 week, 3 month and 1 year follow-up.

Group	TOF group	T1 group	N group	*P*-value
One week				0.361
Complete relief	35 (72.9%)	29 (60.4%)	31 (63.3%)	
Partial relief	6 (12.5%)	12 (25%)	7 (14.3%)	
No relief	7 (14.6%)	7 (14.6%)	11 (22.4%)	
Lost to follow up	0 (0%)	0 (0%)	0 (0%)	
Three month				0.919
Complete relief	31 (64.6%)	28 (58.3%)	26 (53.1%)	
Partial relief	11 (22.9%)	12 (25%)	10 (20.4%)	
No relief	2 (4.2%)	7 (14.6%)	8 (16.3%)	
Lost to follow up	4 (8.3%)	1 (2.1%)	5 (10.2%)	
One year				0.288
Complete relief	27 (56.3%)	28 (58.3%)	28 (57.1%)	
Partial relief	4 (8.3%)	6 (12.5%)	7 (14.3%)	
No relief	4 (8.3%)	4 (8.3%)	2 (4.1%)	
Lost to follow up	13 (27.1%)	10 (20.8%)	12 (24.5%)	

*Values are presented as number (%).

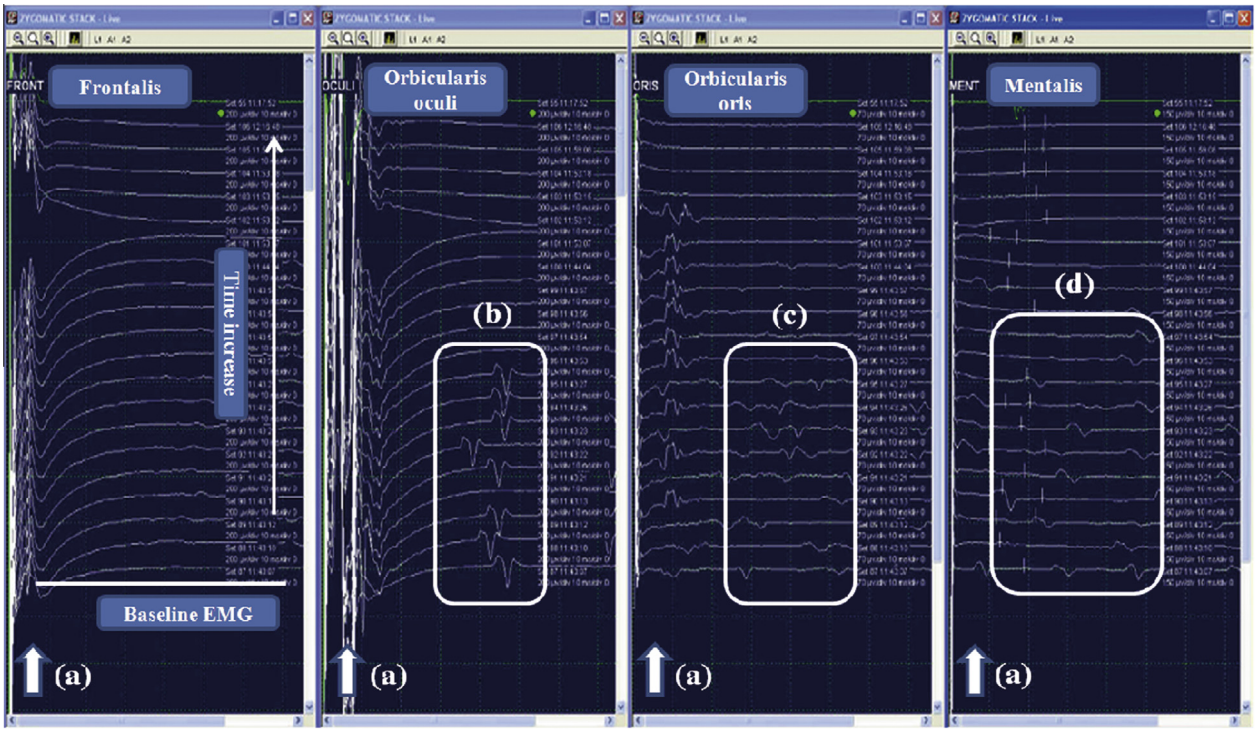

Fig. 5. Stack wave image of an intraoperative facial EMG of a patient in the N group. After one electrical stimulation of the zygomatic branch (a), spontaneous EMG activity was detected in the oculi (b), oris (c), and mentalis (d) muscles. These spontaneous EMG waves disappeared after increasing the level of NMB.

have concluded that a T1/Tc target of 50% is optimal for facial nerve EMG monitoring (Cai et al., 2009; Kizilay et al., 2003). For these reasons, we regarded two counts of TOF and a T1/Tc ratio of 50% as ideal partial NMB targets for facial nerve EMG.

In our study, the success rate for LSR monitoring was significantly lower in the TOF group (66.7%); however, no significant differences were observed between the T1 (91.7%) and N (95.9%) groups. The LSR was not detected in two patients in the N group, even without the use of additional neuromuscular blocking agents. Neves et al. (2009) suggested that a preoperative injection of botulinum toxin could abolish the LSR. In their study, no additional neuromuscular block drug was used, with the exception of a single dose of short-acting neuromuscular block drugs for intubation; moreover, the success rate of LSR monitoring in that study was just 75% (Neves et al., 2009). Since the incidence of patients with preoperative botulinum toxin injections was 25% in our T1 group, the success rate of this group seems to be quite clinically acceptable.

Several reports have shown that facial muscles are less sensitive to nondepolarizing NMB agents than hypothenar muscles (Abdulatif and el-Sanabary, 1997; Kizilay et al., 2003; Kong et al., 2007; Rimaniol et al., 1996). This decreased sensitivity may allow greater motor responses from the facial nerves, compared with ulnar nerves under the same level of NMB. Thus, the use of partial NMB should theoretically allow reliable intraoperative facial nerve EMG monitoring. Accordingly, we found that the success rate of LSR monitoring under partial NMB in the T1 group was not different than the rate of LSR monitoring in the absence of NMB (group N).

In our study, a neurophysiologist continuously monitored free-run EMG activity from the facial nerves, along with the LSR, during microscopic surgery. In some cases, such as when spontaneous EMG signals or excessive EMG noises after stimulation disrupted LSR monitoring (Fig. 5), the neurophysiologist requested that the dose of the NMB drug or the anesthetic depth be increased. The incidence of spontaneous free-run EMG activity was significantly higher in the N group, compared with the TOF and T1 groups. The neurophysiologist requested a continuous infusion of neuromuscular blocking drugs, four cases in the N group due to interference in the LSR monitoring because of spontaneous free-run EMG activity; LSR monitoring became easier after the application of partial NMB in two cases (Fig. 5). These results suggest that partial NMB could be superior to the absence of NMB during LSR monitoring.

Theoretically, it can be assumed that partial NMB can attenuate the amplitude of EMG and also decrease the ability of detecting neurotonic discharges from facial nerve injury. However, there was no significant nerve injury, and the incidence of successful LSR monitoring was not different between T1 and N group. In addition, partial NMB did not impair surgical outcome until at one year. On the contrary, spontaneous EMG activity interfered LSR monitoring in some patients. It seems that the use of NMB influenced on the effectiveness of LSR monitoring, but it did not affect the clinical outcomes.

Contrary to our expectations, spontaneous patient movement during surgery was not significantly different between the three groups (Table 2). However, all movements occurred either when the position of the patient was changed, or during a surgical incision. We increased the TCI doses of propofol and remifentanil upon the occurrence of spontaneous patient movement. No further patient movements occurred after increasing the anesthetic depth. Moreover, no incidences of movement during microscopic surgery were noted.

This study did have several limitations. First, we used continuous infusion of neuromuscular drugs to four patients in N group. The neurophysiologist requested to use the NMB due to excessive spontaneous EMG activity, and we did not want to disrupt the LSR monitoring because of our study. In all four cases, it was considered as monitoring success, since our success criteria of LSR monitoring does not include the absence of spontaneous EMG activity. It did not affect the primary end point of this study. Second, the mean remifentanil TCI dose was significantly higher in the N group compared with the TOF and T1 groups. As we adjusted the remifentanil dose according to the patient blood pressure and mean BIS value was not different between groups, the depth of anesthesia seems not to be different between groups. Furthermore, as remifentanil has not been reported to influence LSR monitoring, the difference in remifentanil dose would not affect the outcome variable. Third, we used adductor pollicis brevis muscle for measuring the degree of muscle relaxation. Ideally, if the facial muscles were used, it could be more helpful for the facial nerve EMG. However, it would be difficult to measure TOF amplitude with facial muscles because frequent muscle contraction by electric stimulation can interrupt both the surgery and LSR monitoring. It seems difficult for anesthetist to conduct quantitative intraoperative NMB monitoring using facial muscles in this surgery.

In conclusion, the success rate of LSR monitoring did not differ between patients under partial NMB with a T1/Tc target of 50% and patients not under any NMB. In some patients, spontaneous free-run EMG activity interfered with LSR monitoring; this was especially problematic for patients not under any kind of NMB. Thus, our data indicate that partial NMB allows good recording of LSR with same outcome as surgery without NMB, and reduced spontaneous EMG activity.

Conflict of interest

None of the authors have potential conflicts of interest to be disclosed.

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References

Abdulatif M, el-Sanabary M. Blood flow and mivacurium-induced neuromuscular block at the orbicularis oculi and adductor pollicis muscles. Br J Anaesth 1997;79:24–8.
Cai YR, Xu J, Chen LH, Chi FL. Electromyographic monitoring of facial nerve under different levels of neuromuscular blockade during middle ear microsurgery. Chin Med J (Engl) 2009;122:311–4.
Costa J, Espirito-Santo C, Borges A, Ferreira JJ, Coelho M, Moore P, et al. Botulinum toxin type A therapy for hemifacial spasm. Cochrane Database Syst Rev 2005;25(1):CD004899.
Dahaba AA, von Klobucar F, Rehak PH, List WF. The neuromuscular transmission module versus the relaxometer mechanomyograph for neuromuscular block monitoring. Anesth Analg 2002;94:591–6.
Fernandez-Conejero I, Ulkatan S, Sen C, Deletis V. Intra-operative neurophysiology during microvascular decompression for hemifacial spasm. Clin Neurophysiol 2012;123:78–83.
Ho LC, Crosby G, Sundaram P, Ronner SF, Ojemann RG. Ulnar train-of-four stimulation in predicting face movement during intracranial facial nerve stimulation. Anesth Analg 1989;69:242–4.
Kim CH, Kong DS, Lee JA, Kwan P. The potential value of the disappearance of the lateral spread response during microvascular decompression for predicting the clinical outcome of hemifacial spasms: a prospective study. Neurosurgery 2010;67:1581–7 [discussion 7–8].
Kim WH, Lee JJ, Lee SM, Park MN, Park SK, Seo DW, et al. Comparison of motor-evoked potentials monitoring in response to transcranial electrical stimulation in subjects undergoing neurosurgery with partial vs no neuromuscular block. Br J Anaesth 2013;110:567–76.
Kizilay A, Aladag I, Cokkeser Y, Miman MC, Ozturan O, Gulhas N, et al. Effects of partial neuromuscular blockade on facial nerve monitorization in otologic surgery. Acta Otolaryngol 2003;123:321–4.
Kong DS, Park K, Shin BG, Lee JA, Eum DO. Prognostic value of the lateral spread response for intraoperative electromyography monitoring of the facial musculature during microvascular decompression for hemifacial spasm. J Neurosurg 2007;106:384–7.
Kothbauer KF. Intraoperative neurophysiologic monitoring for intramedullary spinal-cord tumor surgery. Neurophysiol Clin 2007;37:407–14.
Lee JA, Jo KW, Kong DS, Park K. Using the new clinical grading scale for quantification of the severity of hemifacial spasm: correlations with a quality of life scale. Stereotact Funct Neurosurg 2012;90:16–9.
Lennon RL, Hosking MP, Daube JR, Welna JO. Effect of partial neuromuscular blockade on intraoperative electromyography in patients undergoing resection of acoustic neuromas. Anesth Analg 1992;75:729–33.
Miller LE, Miller VM. Safety and effectiveness of microvascular decompression for treatment of hemifacial spasm: a systematic review. Br J Neurosurg 2012;26:438–44.
Minahan RE, Mandir AS. Neurophysiologic intraoperative monitoring of trigeminal and facial nerves. J Clin Neurophysiol 2011;28:551–65.
Moller AR, Jannetta PJ. Hemifacial spasm: results of electrophysiologic recording during microvascular decompression operations. Neurology 1985a;35:969–74.
Moller AR, Jannetta PJ. Microvascular decompression in hemifacial spasm: intraoperative electrophysiological observations. Neurosurgery 1985b;16:612–8.
Moller AR, Jannetta PJ. Monitoring facial EMG responses during microvascular decompression operations for hemifacial spasm. J Neurosurg 1987;66:681–5.
Neves DO, Lefaucheur JP, de Andrade DC, Hattou M, Ahdab R, Ayache SS, et al. A reappraisal of the value of lateral spread response monitoring in the treatment of hemifacial spasm by microvascular decompression. J Neurol Neurosurg Psychiatry 2009;80:1375–80.
Rimaniol JM, Dhonneur G, Sperry L, Duvaldestin P. A comparison of the neuromuscular blocking effects of atracurium, mivacurium, and vecuronium on the adductor pollicis and the orbicularis oculi muscle in humans. Anesth Analg 1996;83:808–13.
Sekula Jr RF, Bhatia S, Frederickson AM, Jannetta PJ, Quigley MR, Small GA, et al. Utility of intraoperative electromyography in microvascular decompression for hemifacial spasm: a meta-analysis. Neurosurg Focus 2009;27:E10.
Sindou MP. Microvascular decompression for primary hemifacial spasm. Importance of intraoperative neurophysiological monitoring. Acta Neurochir (Wien) 2005;147:1019–26 [discussion 26].
Sloan TB. Muscle relaxant use during intraoperative neurophysiologic monitoring. J Clin Monit Comput 2013;27:35–46.
Sloan TB, Heyer EJ. Anesthesia for intraoperative neurophysiologic monitoring of the spinal cord. J Clin Neurophysiol 2002;19:430–43.

Wang X, Thirumala PD, Shah A, Gardner P, Habeych M, Crammond DJ, et al. Effect of previous botulinum neurotoxin treatment on microvascular decompression for hemifacial spasm. Neurosurg Focus 2013;34:E3.

Yamashita S, Kawaguchi T, Fukuda M, Watanabe M, Tanaka R, Kameyama S. Abnormal muscle response monitoring during microvascular decompression for hemifacial spasm. Acta Neurochir (Wien) 2005;147:933–7 [discussion 7–8].